Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

The vulnerabilities found were passwords that are weak or shared between people, no second step of verification at login, and firewall rules that were not kept up to date. Three hardening measures that address them are:

1. Multifactor authentication (MFA), which asks the user for more than one proof of identity, for example a password plus a fingerprint, an ID card or a code.

2. A password policy, with rules on length and allowed characters, a ban on sharing passwords, and a limit on failed login attempts (for example, the account is locked after five wrong tries).

3. Regular firewall maintenance, which means checking the rules and updating them so they match current threats.

Part 2: Explain your recommendations

MFA adds a layer on top of the password. A brute force attack alone is no longer enough, because the attacker also needs the second factor. It also makes sharing a password pointless, since the other person would not have the second factor.

A password policy makes the network harder to get into. Locking an account after a few failed attempts stops brute force attacks, and longer, more complex passwords that cannot be reused or shared make guessing much harder. People will still make mistakes, but the policy removes the most common ones.

Firewall maintenance keeps the allowed and denied traffic rules current. Suspicious sources go on a deny list, and the rules are reviewed after every security event, especially if it let suspicious traffic in. This helps against DoS and DDoS attacks.